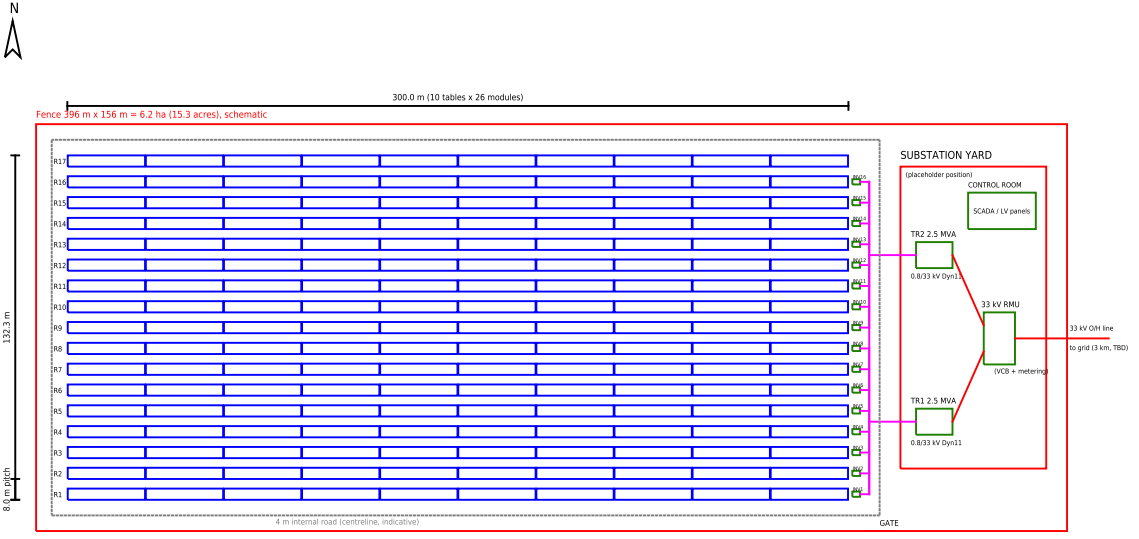
5 MWp SOLAR PV PLANT: PRELIMINARY PLANT LAYOUT (PLAN)

Fixed tilt 20 deg, azimuth 0 deg (south), pitch 8.0 m, GCR 58%. Schematic: inverter, transformer and yard symbols are enlarged / placeholders.



KEY DATA

DC 4,994.6 kWp (8,840 x 565 Wp bifacial) / AC 4,000 kW (16 x SG250HX)
17 rows x 10 tables = 170 tables (2 portrait x 26 modules = 52 modules each)
340 strings of 26 modules (2 strings per table)
Array footprint 300.0 m x 132.3 m = 3.97 ha (excl. roads, yard, fence)
2 x 2.5 MVA 0.8/33 kV transformers: TR1 takes INV1-8, TR2 takes INV9-16
PVsyst VC4: 7,078 MWh/yr; 6,917 MWh/yr at 33 kV point after AC losses
Strings-to-inverter assignment and cable routes are indicative (not yet designed)

LEGEND

- PV table (2 portrait modules x 26)
- Inverter / transformer / RMU (schematic)
- LV AC cable (240 mm2 Al, 800 V)
- 33 kV link
- Internal road (indicative)
- Fence line / yard boundary

0 — 50 m

PROJECT	5 MWp ground-mount solar PV plant, West Bengal (resource site: Kolkata)
DESCRIPTION	Plant layout (plan)
DRAWING NO.	SPV5-DWG-01-Layout-R0
SCALE	As plotted on A3 (see scale bar)
DRAWN / DATE	Sagnik Kumar Nath / 28-10-2026
STATUS	PRELIMINARY, NOT FOR CONSTRUCTION